

Maninderpal Singh

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EDUCATION

Punjabi University

Department of Statistics — B.Sc. in Computer Science, Statistics & Mathematics

Percentile 89.8%

Jul 2019 – Jul 2022

EXPERIENCE

Instahyre

SDE 1

Sep 2023 – Present

New Delhi, India

- **Module Ownership:** Owned 4 platform modules at different stages, taking responsibility for feature development, code reviews, performance, monitoring & incident response
- **Platform Reliability:** Did RCA, resolved functional & performance issues across modules (650+ tickets overall), cutting recurring bugs by 40% in 2 quarters and boosting the legacy system's stability, reducing tech debt.
- **Product Collaboration:** Partnered closely with Product Managers and cross-functional teams to design and implement features across in-progress and existing modules & contributing to iterative product improvements

Buyceps

Software Engineer

Dec 2022 – Sep 2023

Mumbai, India

- **Backend Leadership:** End-to-end ownership for backend development of Buyceps Marketplace, implementing unified authentication, order/shipment workflows, and end-to-end reliability
- **Search System:** Built an Algolia powered product browser with optimized DB indexing and internal UI tools for fast, efficient search
- **Performance Optimization:** Optimized major GraphQL resolvers and Python/Django REST APIs, reducing N+1 queries and cutting SQL calls by 75%.

Studypaq

Software Engineer

Oct 2022 – Dec 2022

California, United States

- **Security & Performance:** Discovered and fixed a critical vulnerability in a paywall protected module
- **API Design & Data Modeling:** Spearheaded the designing and developing core APIs and architecture
- **Server performance:** Improved response time of high-traffic endpoints by 47%, reducing SSR latency

PROJECTS

Hunter - Online contests hosting platform (SaaS)

Jan 2022

Repository, Live

- **Architecture:** Engineered a full-stack SaaS using Node.js (Express, TypeScript), Angular, MySQL
- **Code Execution Engine:** Created my own distributed remote execution service using GoLang, RabbitMQ
- **Deployment:** Deployed using NGINX on Azure VM, later migrated infrastructure to Oracle Cloud and PlanetScale; later moved to bare-metal homelab; later moved again to Oracle on ARM based VM

Showdown - Distributed Remote Code Execution Engine

Aug 2024

Repository, Documentation, DockerHub

- **HTTP server:** Built a high-throughput scalable remote code execution engine and APIs in Go
- **Containerization:** Using ISOLATE (open source project) for secure, containerized execution of untrusted code
- **Distributed:** Implementation with a manager-worker architecture using RabbitMQ

DinoPlusPlus - A Terminal Based C++ Game (ASCII Graphics)

Apr 2021

Repository

- **Gameplay:** Programmed the entire game play mechanics, menu system & animations without any specialized console game engine or code library
- **Frame desynchronization:** Fixed by caching the frame until fully computed and then drawing all at once
- **Collision detection:** Implemented proper edge detection of any object colliding, render time

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, GoLang, C/C++, Bash

Tools/Frameworks: Django, Angular, REST APIs, Git, Express.js, Node.js, GraphQL, Agentic AI/LLM Automation

Data: MySQL, ElasticSearch, Redis, RabbitMQ, PostgreSQL, MongoDB

Cloud & DevOps: Docker/Containerization, NGINX, Scripting, AWS, Microservices

Core Competencies: Scheme Design, Distributed Architecture, Object-Oriented Design (OOP), Query Optimization